

AI Literacy and Reported Usage with General Knowledge, TRI, and Autonomy (#159548)

Author(s)

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Pre-registered on: 2024/01/25 - 11:45 AM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

We are examining whether AI literacy is associated with receptivity toward AI as measured by usage of AI tools over the past 6 months. Specifically, we expect lower AI literacy to be associated with higher receptivity to AI.

3) Describe the key dependent variable(s) specifying how they will be measured.

AI receptivity will be measured by the following:

In the last six months, how often have you used...

- (i) ☐ A digital image generator (e.g., DALL-E, Midjourney,...)
- (ii) ☐ An AI-powered productivity tool for note taking, scheduling, inbox management (e.g., Zapier),
- (iii) ☐ An AI-powered design service to create a website (e.g., Canva)
- (iv) ☐ An AI-powered health app to meditate, monitor sleep (e.g., Headspace, Sleep.ai)
- (v) ☐ An AI-tool as a writing assistant (e.g., ChatGPT)

all on 5-point scales: 1- Never, 2-Once or twice, 3-Occasionally (more than once or twice but not more than once a month) 4-Frequently (more than once a month), 5-On a weekly basis. We will average responses to these 5 items into an index of AI usage.

4) How many and which conditions will participants be assigned to?

There are no conditions. Our independent variable will be participants' AI literacy as measured using a 17-item multiple choice test. The sum of the correct responses will serve as their AI literacy level.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will regress AI usage on AI literacy.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We include one instructional manipulation check (IMC; "Please indicate you are taking the time to read the instructions carefully. This is not a trick question. What is 12-4? Please use a numeric value only (no letters)."). This will be asked at the beginning of the survey and participants who fail this question will not complete the survey.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will post the survey online for 400 respondents.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will measure general knowledge with 20 multiple choice questions for each of the following four topics (5 questions each): History, Biology, Geography, and Literature. We will sum the correct responses to form a general knowledge measure.

We will measure Technology Readiness Index (TRI) using the 10-item scale from Victorino et al. 2009. We will average responses into an index capturing technology readiness perceptions (reverse coding the items assessing discomfort and insecurity).

We will collect general attitudes towards self-reliance versus dependence on others via motivation for autonomy (from Dahl and Moreau 2007): "To what extent do the following statements describe you? (i) I want to be free to make my own choices (ii) I want to express myself (iii) I don't want to be controlled (iv) I don't want to feel pressured 1- Not at all 7 – Extremely. We will average responses to these items into an index capturing self-reliance.

We will also measure demographic information (gender, age, income).

For each of the above measures, we will explore if they are significantly correlated with AI literacy and AI usage. For any and all measures that are significantly related to both AI literacy and AI usage, we will repeat the analysis specified under (5) controlling for these measures.